



Software Engineer



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## PERSONAL STATEMENT

My goal is to leave the world a better place than I found it. I plan to bring that goal into reality by creating technology that improves the quality of life. These goals have motivated my passion for software engineering and the open-source community as a tool for sharing knowledge. My internship at NIWA involved collaborating with scientists and physicists to publish their research on our oceans and atmosphere to a global audience. Software serves as a medium to explore my scientific curiosity and contribute to meaningful change.

## PUBLICATIONS

1. Wood, J., Nguyen, B., Xue, B., Zhang, M., & Killeen, D. (2025). Hook, line, and spectra: machine learning for fish species identification and body part classification using rapid evaporative ionization mass spectrometry. *Intelligent Marine Technology and Systems*, 3(1), 1-15.
2. Wood, J., Nguyen, B. H., Xue, B., Zhang, M., & Killeen, D. (2022, December). Automated fish classification using unprocessed fatty acid chromatographic data: a machine learning approach. In *Australasian Joint Conference on Artificial Intelligence* (pp. 516-529). Cham: Springer International Publishing.

## EXPERIENCE

**PhD Student** - 2022 to Present - *Victoria University of Wellington*, Wellington, NZ

- Awarded a Wellington Doctoral Scholarship by the Engineering faculty for excellence in undergraduate studies.
- Awarded an MBIE Doctoral Grant by the Engineering faculty.
- Conducted marine biomass analysis of mass spectrometry data using deep learning, evolutionary computation, and machine learning techniques.
- Utilized sklearn, PyTorch, and DEAP to construct advanced machine learning tools.
- Applied machine learning in chemistry with Rapid Evaporative Ionization Mass Spectrometry.
- Developed models to detect fish species, body parts, oil, and cross-species contamination with high accuracy.
- Published a journal paper in *Intelligent Marine Technology and Systems* detailing this work.
- Published a research paper at AJCAI 2022 detailing this work.

**Research Assistant** - 2021 to 2022 - *Victoria University of Wellington*, Wellington, NZ

- Awarded a Summer Research Scholarship by the Engineering faculty.
- Generated a comprehensive database of sound fields in rooms using the Image-source method.
- Developed Conditional Neural Processes using PyTorch to accurately interpolate sound fields.
- Authored a research paper summarizing experiments and their analysis.

**Tutoring** - 2021 - Present - *Victoria University of Wellington*, Wellington, NZ

- Provided academic support for COMP309 - Machine Learning (2023).
- Tutored SWEN304 - Databases and System Engineering (2023).
- Guided students in ENG301 - Engineering Project Management (2022).
- Supported ENGR401 - Professional Practice (2022).
- Assisted with ENGR302 - Engineering Project Management (2023).
- Instructed SWEN325 - Software Development for Mobile Platforms (2022, 2021).

**Web and Content Intern** - 2021 - *Wellington Chamber of Commerce*, Wellington, NZ

- Gained a strong understanding of business and legislation, developing a toolkit for successful business operations.
- Implemented data visualization techniques to create engaging and interactive tools for business and marketing analysis.
- Performed web development using the iMIS Engagement Management System, maintaining and developing dynamic components from a database.
- Collaborated with marketing and design teams using agile methodology to deliver effective products meeting user requirements.

#### **Software Intern** - 2020 to 2024 - *N/WA*, Wellington, NZ

- Implemented an end-to-end best practice OGC Standard software through multi-disciplinary collaboration.
- Developed web services utilizing an open-source technology stack to query a database on a Linux cloud server.
- Communicated progress in weekly meetings and documented work for non-technical users using best practice agile methodology.
- Successfully worked remotely during the COVID-19 pandemic, utilizing industry-standard tools (e.g., Microsoft Teams for communication and Pulse Secure for SSH).

#### **Front of House** - 2018 to 2019 - *St. Johns Bar and Eatery*, Wellington, NZ

- Organized and led a team to work efficiently in a fast-paced customer service environment.
- Delivered exceptional value to customers and maintained transparency regarding operational challenges.

#### **Back of House / Front of House** - 2012 to 2018 - *Mac's Brewery*, Wellington, NZ

- Managed high-pressure situations in a fast-paced work environment, including risk management and negotiations with disorderly customers.
- Employed empathetic and active listening skills to de-escalate confrontations effectively.

## **EDUCATION**

#### **Doctor of Philosophy (Artificial Intelligence)** - 2022 to Present - *Victoria University of Wellington*

- Awarded an **MBIE Doctoral Grant**.
- Awarded the prestigious **Wellington Doctoral Scholarship** for excellence in undergraduate studies.

#### **Bachelor of Engineering (Software Engineering)** - 2016 to 2022 - *Victoria University of Wellington*

- Completed subjects providing a strong academic background for a full-stack developer, including: **Industry Research Project, Human-Computer Interaction, Deep Neural Networks, Compiler Engineering**.
- Served as **Class Representative** for SWEN430 Compiler Engineering, acting as a liaison between students and faculty for course-related issues.
- Volunteered as a **Note-Taker**, ensuring university accessibility to a wider audience through information dissemination.
- Received the **Victoria Excellence Scholarship** for excelling in secondary school studies.

*N.B The Head of School's commentary on my grades is attached to this document.*

#### **NCEA Level 3** - 2011 to 2015 - *Rongotai College*

- Core subjects provided a strong foundation and fostered a passion for Computer Science: **Computers, Physics, Calculus, English, Graphics, Music**.
- Developed and honed public speaking and presentation skills through extra-curricular roles: **Prefect, UN Youth Ambassador, Jazz Band, Production Band, Debating**.
- Achieved **Scholarship English** - an award for university-level written communication skills given to secondary school students.

## PROJECTS

### Fish Oil Analysis

A machine learning project collaborated with industry partner New Zealand Plant & Food. Performed classification and feature selection on Gas Chromatography data from fish oil. Achieved near-perfect accuracy (98.3%) in classifying fish species from tissue samples using a Support Vector Machine model. This research significantly impacts food science by enabling the automation of a laborious manual process in factory settings.

### Data Ingestion

Ingested multiple datasets into the NZODN portal, an open-source technology stack implementing OGC standards. Utilized tools such as GeoServer for web map services and GeoNetwork as a CMS for metadata catalogues. Configured the software to integrate with a Postgres database storing geospatial data.

### Resource Portal

Developed an automated workflow for the Wellington Chamber of Commerce's iMIS Engagement Management System (EMS) to digitize paper-based documents. Transformed these documents into responsive and engaging web pages, streamlining content management.

### Advent of Code

Solved programming puzzles from an advent calendar using Haskell, a functional programming language. Leveraging Haskell's immutability, solutions were developed with reduced likelihood of errors, leading to faster problem-solving and accurate submissions after thorough test case verification.

## TECHNICAL TOOLS

**Scripting:** Demonstrated general knowledge of fundamental principles and tradeoffs across a breadth and depth of programming languages, encompassing both Functional and Object-oriented paradigms, and a mix of High and Low-level languages. **React, Next.js, Rust, Node, GraphQL, Tailwind CSS, Deno, Haskell, Java, C++, C, Ruby, Unreal Engine**

**Machine Learning:** Possess practical experience in implementing end-to-end machine learning pipelines to extract business knowledge from real-world datasets. Proficient in using scripting languages for data preprocessing, exploratory data analysis, model training and testing, and data visualization. **Python, PyTorch, scikit-learn, Pandas, Numpy, Hugging Face**

**Databases:** Hold academic and practical knowledge in both SQL and NoSQL database systems. Experienced in creating and utilizing databases with cloud services for full-stack applications. **PostgreSQL, GeoSever, GeoNetwork, Supabase, Firebase, AWS, Docker**

**Workflow:** Proficient with Git for version control and agile documentation of development. Experienced in leveraging Git-based platforms to facilitate asynchronous and remote collaboration within development communities. **Git, Gitlab, Github, GitBucket, Jira, LaTeX, Markdown, PlantUML**

## WORKSHOPS AND CONFERENCES

### New Zealand Seafood Conference 2024

A two-day conference on the New Zealand seafood industry, featuring talks from field experts. Provided a comprehensive overview of the seafood industry and unique perspectives from its professionals.  
Auckland, New Zealand - Attended in person  
7-8 August 2024

### AJCAI 2022

The 35th Australasian Joint Conference on Artificial Intelligence (AJCAI) held hybrid mode in Perth. This flagship conference for the Australasian AI community served as an annual event to advance communication between academic researchers and industry AI practitioners.  
Perth, Australia  
Dec 5-8, 2022

### GECCO 2022

The Genetic and Evolutionary Computation Conference (GECCO) presents the latest high-quality results in genetic and evolutionary computation since 1999. This conference offered an invaluable opportunity to learn about state-of-the-art evolutionary computation.  
Boston, USA - Attended Virtually  
July 9-13, 2022

### EvoStar 2022

The Leading European Event on Bio-Inspired Computation. This conference provided an excellent opportunity to learn

about evolutionary computation research and its applications.  
Madrid, Spain - Attended Virtually  
20-22 April 2022

### **MAXAR Hackathon 2021**

Participated in a 24-hour GIS + AI hackathon. Our team developed a sea-level-rising visualization web application and a supervised object detection algorithm to identify houses from satellite imagery along coastlines. Maxar is a space technology company specializing in satellite and spacecraft component design and manufacturing.

### **WOSSAT**

Attended the Wellington Open Source Show and Tell (WOSSAT), a monthly meetup hosted by Catalyst. This meetup showcases notable new technologies, and hobby projects, and promotes open-source advocacy, emphasizing open standards and open data to reduce barriers to technology access.

### **FOSS4G SoTM**

Attended a local hub for an international event covering cutting-edge open-source GIS software. A key takeaway from the conference was the critical importance of ethical considerations in software development, highlighted by a former MapBox developer whose software was used for drone strikes.

### **Python Data Ingestion**

Attended by data scientists and software developers, this workshop focused on using Python for Scientific Computing. Explored the Anaconda environment for Python development and package management, learning its utility in replicating Python environments across machines. Highlighted Jupyter notebooks as an important tool for Literate Programming, merging documentation and codebases.

### **Databases 101**

Engaged in discussions comparing RDBMS vs. NoSQL databases, exploring tradeoffs between software quality attributes and risks of vendor lock-in with proprietary cloud software (e.g., AWS, Azure, Firebase). Gained valuable insights into understanding end-user goals for database creation and maintenance.

## **HOBBIES**

### **Debating**

Public speaking, politics, philosophy, UN Youth Ambassador

### **Sciences**

CS, AI, Robotics, Astrophysics, Psychology

### **Guitar**

Classical, Jazz, Blues, Classic Rock

### **Sports**

Canoe Polo, Table Tennis

## **REFERENCES**

**Andrea Mari** - Research Software Engineer at NIWA

Andrea is my mentor at NIWA. He introduced me to professional software development processes, including Jira and agile methodologies. I have greatly benefited from his industry expertise as a software developer.

**Bing Xue** - PhD Supervisor, Deputy Head of School Research & International, School of Engineering and Computer Science, *Victoria University of Wellington*

Bing is one of my PhD supervisors. She supervised my industry project involving marine biomass analysis using machine learning techniques on rapid evaporative ionization mass spectrometry.

**Miniruwani Samrakoon** - Co-President, Postgraduate Students Association at *Victoria University of Wellington*

Miniruwani was my tutor for a 300-level Engineering group project. This collaboration involved agile methodologies to develop a web application that provided a Mission Control interface for a Rocket.

*N.B. Contact information is available upon request.*



VICTORIA UNIVERSITY OF  
**WELLINGTON**  
TE HERENGA WAKA

**SCHOOL OF ENGINEERING AND COMPUTER SCIENCE**

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26 November 2021

Jesse Wood

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Wellington 6011

Tēhā Koe Jesse

On behalf of the School of Engineering and Computer Science, I would like to offer you a warm congratulations on achieving the grades SWEN422: A+ and SWEN430: A-

It gives the school great pride when students achieve academic success at this level. I am sure it has entailed a good deal of hard work on your part and is well deserved.

My colleagues and I will be happy to give advice on opportunities at both undergraduate and postgraduate levels and we look forward to supporting your future academic success.

We hope that you have a relaxing break over the summer holidays.

Ngā mihi

Dr Craig Watterson  
**Head of School**



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13 November 2020

Jesse Wood

7 Humber St, Island Bay

Wellington 6011

Tēhā Koe Jesse

On behalf of the School of Engineering and Computer Science, I would like to offer you a warm congratulations on achieving the grades COMP309: A, ENGR302: A and SWEN325: A

It gives the school great pride when students achieve academic success at this level. I am sure it has entailed a good deal of hard work on your part and is well deserved.

My colleagues and I will be happy to give advice on opportunities at both undergraduate and postgraduate levels and we look forward to supporting your future academic success.

We hope that you have a relaxing break over the summer holidays.

Ngā mihi

Dr. Stuart Marshal  
**Head of School**